Essence C

A new programming language to advance link-time optimizations by faciliating the continuum of static and dynamic modules.

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Declaration

I hereby certify that this material, which I now submit for assessment on the program of study as part of the B.Sc. Computational Thinking qualification, is entirely my own work and has not been taken from the work of others - save and to the extent that such work has been cited and acknowledged within the text of my work.

I hereby acknowledge and accept that this thesis may be distributed to future final year students, as an example of the standard expected of final year projects.

Signed: _____ Date: _____

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A massive thanks to Dr. Phil Maguire, friends, and peers for inflicting levels of motivation I might never reciprocate.

Abstract

The logic of computer software is primarily expressed using programming languages, but the balance between versatile and efficient programming languages remains a major challenge. Changes to the execution environment of a program can impact performance, or worse, cause incompatibilities that require major refactoring. Developers often maintain multiple branches of a codebase to keep performance to acceptable levels. One particular difference between execution environments is whether individual modules are statically linked at compile-time or dynamically resolved at runtime. In this research project, we propose a prototype programming language called "Essence C" that better supports the continuum of static and dynamic modules, by allowing the programmer to write code once that is agnostic to the linking strategy. By taking advantage of LLVM's intermediate representation, it is possible to advance the state-of-the-art of link-time optimizations across module boundaries.

Chapter 1 – Introduction

1.1 Overview

1.2 Motivation

Software development is becoming very popular, with graduates now unable to find jobs due to a massive spike in educational resources allocated to this field. With the use of LLM agents gaining traction, keeping a codebase clean may be harder than ever before. There is now serious competition in programming language design due to two reasons. Firstly, an LLM's ability to navigate a codebase effectively will depend on the structure imposed directly by the language, given the reduced structural input from the programmer themself. Techniques that minimize duplicated effort in a codebase is expected to minimize context rot and reduce the agentic workflow bills. Secondly, agents are at risk of optimizing their development in highly trained-on environments (e.g. JavaScript, web-based, and React.js environments), leaving less-prevalent low-level languages behind unless the agent can learn from on-the-fly tool calls. Highly structured languages with a superb type system, high cohesion of language concepts, and excellent documentation may make up for rhe lack of training. An effective module system alongside other novel language features has the potential to transform the way humans and agents reason about their software.

1.3 Methodology

Thesis structure, and a resesarch strategy.

To address the objectives of this research, a design-science methodology is employed with a focus on continued interation and evaluation of "Essence C". This approach was selected because the project is concerned with the implementation of a novel programming language that advances the flexibility and performance of modules in software development, and thus requires a design and implementation process that is informed by ongoing evaluation.

The structure of this thesis is organized to reflect this methodology. Chapter 2 reviews existing literature.

1.4 Success Criteria

Success is evidenced by the following deliverables:

(1) A functional prototype compiler that implements the designed module system;

(2) Documentation of the module system and other core language features;

(3) A simple IDE extension that provides syntax highlighting to the programmer;

(4) A body of research into the continuum of static and dynamic modules in software development, with kernel development as a case study;

(5) An implementation of link-time optimization techniques that leverage the designed module system;

(6) An evaluation of the effectiveness of these optimizations through the benchmarking of sample programs;

(7) A monorepo containing extensive git history, demonstrating ongoing development progress;

(8) A final written report (that you are reading) detailing the research, design, implementation and evaluation of the project.

1.5 Use of Generative AI

Two notable GenAI tools have been used to assist with the research and development of this project.

(1) The first is t3.chat, a web application that consolidates numerous large language models into a streamlined chat interface **t3chat**. The user can experiment with different models to understand which models are best suited for the task at hand. Both research and programming tasks have benefited from the use of this tool. The tool was primarily used to steer the work in the right direction, and to identify the concepts, libraries and resources through which further research could be conducted. For transparency, all LLM prompts have been included in **Appendix C**. It was

found that the tool provided a simple JSON export facility to allow for programmatic filtration of relevant chats.

(2) The second is the tab-complete feature of GitHub Copilot. This tool has encouraged rapid development and iteration through its behaving as a glorified autocomplete. From experience, the tool's context is severely limited, forcing the developer to have genuine knowledge of the codebase and its architecture. This is beneficial for projects that require significant unique input from the developer, in contrast to rehash projects that benefit more directly from LLM assistance.

Some experimentation with GitHub Copilot agent mode was conducted, with notable issues identified. Firstly, the development speed was impeded by slow model response times, lack of funds, and internet connectivity issues. Secondly, there was a great difficulty in referencing contributions made by this tool due to (1) tool calls that modify the codebase directly and (2) significant implicit pollution of the context window as the agent navigates the respository. As such, usage of this tool was suspended.

Chapter 2 – Technical Background

2.1 Literature Review

2.1.1 Static vs dynamic dispatch

Methods of class instances (or equivalent constructs) dispatch calls either statically or dynamically. When the exact implementation of a method is known at compile-time, static dispatch is typically used, involving a direct assembly call to the method. Since the method is known at compile-time, the compiler can use an immediate addressing, which is most efficient. In contrast, when the exact implementation of a method is not decided until runtime (or if the compiler cannot prove otherwise), dynamic dispatch is used, which involves an indirect call through a vtable or similar structure.

2.1.2 Static vs dynamic linking

There is much overlap in the literature between dispatch and linking, but linking focuses on larger units of code, often referred to as modules or libraries, as opposed to class instances. This concept exists in C where there is no inbuilt notion of static vs dynamic dispatch.

When the exact source code (implementation) of a module is known at compile-time, static linking is typically used, involving the direct inclusion of the module's code into the final executable. Then, the compiler can use immediate addressing to invoke functions or access data from this module. In contrast, when the exact source code of a module is not decided until runtime, dynamic linking is used which typically requires a Procedure Linkage Table (PLT) and Global Offset Table (GOT).

However, various techniques are already employed to optimize dynamic linking:

(1) Some architectures use the "relaxation" technique to replace indirection at runtime by modifying the assembly instructions.

(2) PLT Rewriting provided by compilers.

(3) Bespoke replacement (e.g. linux kernel static_call).

However, two primary concerns of interest are still evident:

(1) There is always some level of pointer indirection that cannot be avoided (nvm);

(2) Various techniques such as inlining become impossible due to this indirection.

Terrible PIMPLE idiom. Headers of headers.

2.1.3 Module loading in the Linux kernel

It was found that the potential inefficiencies of dynamically loaded modules were already considered by the Linux kernel developers, and boutique infrastructure was developed to partially mitigate this.

The Linux kernel benefits from their "static_call" system, which uses a combination of C macros, custom compilation/linking steps, and runtime logic to patch function calls to be more efficient. Any calls that may need to be dynamically swapped at runtime can engage with the "static_call" system to improve performance and automate the repetetive assembly logic. The process is summarized below.

Macro system

Each static-call function is implemented using the "DEFINE_STATIC_CALL" macro, which takes in (1) a handle name, and (2) a pointer to the initial implementation for this handle. This macro is used at the top-level. Then, the programmer can use the "static_call_update" macro at any point, taking in the same parameters, to alter the implementation for a particular handle at runtime. Alas, the programmer can use the "static_call" macro to invoke the handle like a traditional function. Here is some example pseudo-code:

```
void implementation_a(int value) { ... }
void implementation_b(int value) { ... }

DEFINE_STATIC_CALL(foo, implementation_a);

void main() {
```

```
  static_call(foo)(27);
  static_call_update(foo, implementation_b);
  static_call(foo)(27);
}
```

The “static_call” macro also uses the inline assembly “.pushsection” and “.popsection” directives to temporarily relocate the code to a unified object section that can be picked up in the custom compilation step.

A crucial takeaway is that the lack of direct language support has forced the need for a custom macro setup. Understandably, programmers in different codebases may not be so adamant to commit resources to such a setup in the same way the Linux kernel developers have.

Custom compilation step (objtool)

The kernel needs to remember where the call sites are when the implementation is swapped at runtime. To achieve this, the kernel includes an extra build step that investigates the aforementioned object section to locate all relevant call sites. It then builds up a table that can be inspected at runtime to ensure no call sites are missed. There are many more intricacies that are not mentioned and have not been studied in too much detail.

If you explore the source code for the Linux kernel, you will come across the static_call_types.h file, containing the following:

```
/*
 * The static call site table needs to be created by external tooling (objtool
 * or a compiler plugin).
 */
struct static_call_site {
    s32 addr;
    s32 key;
};
```

Trampoline solution

The easy solution to alter the implementation of a function at runtime is to use a trampoline (an extra dynamic jump instruction that is resolved at runtime). This is slow and is merely a backup option.

Insert diagram.

Inlining solution

Great news, inlining refers to the trampoline call, not inlining the function body itself.

(Well not so great news, but at least there is more merit to this project).

Spectre

Mention spectre.

2.1.4 Link-time optimizations

Traditionally, many optimizations were performed at the language level, before the program was compiled to object files. For instance, inlining could not be performed at link-time because there was insufficient information in the object files to reason about the semantics of the program. While dedicated link-time optimizations were introduced, their scope was limited for this reason.

However, LLVM IR sits in between the level of abstraction of source code and object files, and maintains higher-level semantics. By keeping the program in this intermediate representation longer (and treating this as modern object files), recent developments have made link-time optimizations more powerful, allowing, for instance, for inlining across module boundaries.

2.1.5 Inlining and IR-level linking

2.2 Technical Material

2.2.1 LLVM intermediatea representation

Chapter 3 – The Problem

3.0.1 Indirect Calls

The core of the problem is removing indirection.

Chapter 4 – The Solution

4.1 Design Elements

4.1.1 Choice of Language

The languages selected for this project were JavaScript and C++. While JavaScript has been my go-to language before classes were even a thing, the primary LLVM API is sadly exposed through C++. Other language bindings exist, but do not appear mature enough. It would not have been smart to gamble a project of this calibre on the stability of such bindings. JavaScript was still suitable for the parsing phase, as my pre-existing parsing library was already written in JavaScript. The speed of compilation is mostly irrelevant, as this does not impact the performance of the compiled artifacts. Thus minimal effort was wasted on such a triviality.

4.1.2 Unification of Modules and Classes

Languages have historically offered various features like classes, modules, packages, and so on, that have much overlap in their functionality. One of the core design decisions was to unify these elements into a single system that consists of a smaller unit called a "map" (which bares some resemblence with Rust- and Go-like structs), and a larger organizational unit called a "module". Most importantly, every module still uses a map for instance-management purposes, and so the same capabilities are available whether or not a program is split into more modules or not. This differs from languages like C, where the choice to have separate compilation units has an impact on the final program [ref]. The decision for this unification was inspired by Rust's philosophy of "zero-cost abstractions" [ref], an acknowledgement that organizational differences in a program that aid with design and iteration should not impact the performance of the final artifact.

4.1.3 Map-Enum Equivalence

Three features in programming languages were found to share an ermegent goal. These are structs (or classes), functions (or methods), and enums (specifically, tagged unions).

Essence C unifies structs and functions into a single feature called a "map". This takes inspiration from Pure Mathematics, where the pattern of notational definitions is often used for multiple purposes – functions ($f(x)$), operations ($x+y$), subscripts (x_n), accessors ($\star_G$). Little special treatment is given to these different cases, as all could be denoted with function-like syntax ($f(x), +(x, y), x(n), \star(G)$).

In Essence C, there are no struct accessors like ".first_name". Rather, a struct is a "map" that takes in a symbol (e.g. ":first_name") as its only parameter and returns the member from its internal storage. These semantics are not so bizarre, as the technique to store data alongside functions (often known as closures) is already commonplace. When a "map" has no virtual methods, its behaving as a struct results in the same low-level data representation.

Insert beautiful diagram here.

Furthermore, tuples are syntactic sugar for maps, adding positional syntax. One notable outcome is that there is no concept of multiple function parameters; instead, there is a single parameter which reuses the existing tuple syntax to achieve multiple parameters. It's maps all the way down.

Next, Essence C unifies maps and enums via the so-called Map-Enum Equivalence technique.

Recall Rust-like enums (tagged unions) where a symbol (:foo_bar) is associated with some data (:foo_bar("hello")).

A syntactical ambiguirty arises. If we write map:foo_bar("hello"), are we calling the map with a symbol (a bare enum) and then calling the resulting map with ("hello"), or, are we calling the map with a fully populated enum? The solution is to unify these two approaches, by demanding that all maps obey map-enum equivalence, that is, that calling a map with a symbol and then another map, is equivalent to calling the map with the fully populated enum. This bears partial resemblance with currying. What's amazing now is that populated enums are really just pre-curried symbols. The vast majority of code written by the programmer now comes from maps and symbols.

4.1.4 Type-Value Unification

Old-fashioned languages like C treat types and values as entirely distinct.

In the traditional sense, a type offers two functions: - It details the runtime representation of the data, i.e. the possible runtime values that may be encoded and how many bits of data are needed. - It provides a constraint to the programmer themself, preventing the programmer from using values that the code is not built for.

It turns out that it may be appropriate for these two intents to diverge. This results in two separate notions of a type:

- **Logical Type:** A set of allowable values. A constraint of what theoretical values are permissible, i.e. subject the programmer's intended business logic. - **Encoding Type:** A bijection between a set of values and a set of bit patterns. This allows types to be designed and abstracted independent of the encoding used.

In this sense, a value is just a singleton logical type.

encoding value vs theoretical value.

both the concept unified, and then the syntax is unified too.

4.1.5 Additional benefit of inlining: interacting with existing languages without indirection.

4.2 Picking of overall solution

4.3 Design

4.3.1 Associativity

In most programming languages, addition and subtraction are treated as unary operations in an additive batch, whereras operations like multiplication and division are instead treated as binary operations in a multiplicative tree. However, it turns out that any associative operation (such as multiplication) can be written in the former form, where operations in the same family (such as division) control the unary effect via an operator that prefixes the operand. As such, a consistent syntax was employed in this language, allowing both the unary and binary approach for all associative operations and their respective intra-family operations. This includes the following families:

- Addition and subtraction (like existing languages);
- Multiplication and division;

- Logical AND;
- Logical OR;

The syntax was designed to follow the crystal/pistol pattern, some terminology devised to mean the following:

- Crystal: A unary batch of operator/operand pairs (an extension of how existing languages handle addition and subtraction).
- Pistol: A binary tree of operations (like how existing languages handle multiplication and division).

For precedence reasons, we have eight separate rules: OR crystal -> OR pistol -> AND crystal -> AND pistol -> additive crystal -> additive pistol -> multiplicative crystal -> multiplicative pistol, with each embedding the next in the hierarchy. Those further to the right of the chain enjoy higher precedence.

More details on this pattern can be gathered by viewing my article on **boundly associative operations**, my article on **making complicated if statements cleaner**, and my YouTube video on **rethinking operation order**.

4.4 Implementation

4.4.1 Mutability

LLVM IR is in SSA form, where registers are immutable. One cannot reuse a register, and must instead create a new register for every mutation. Then, one takes advantage of the phi instruction to pick which register is the correct one at runtime. However, the alloca (stack allocation) model operates under the familiar mindset of mutability, as it works with raw memory addresses. Conveniently, it was observed that the standard mem2reg optimization pass does a fantastic job at converting stack allocations into SSA registers where legal, bypassing the tough development of a custom SSA mutability mechanism. Thus, variables are entirely implemented using stack allocations. This saved an estimated week of development time.

4.4.2 Tooling and Infrastructure

Parser Trace Explorer

Parsing issues were frequently encountered during the development cycle, and manual debugging became infeasible as complexity increased. Hence, a *Parser Trace Explorer* was developed, consisting of a JSON trace logger in the compiler frontend, and a separate web-app tool for exploring these logs. A simple SolidStart + Tanstack Router tech stack was selected for this tool due to existing familiarity with this stack, and because of the straightforward requirements of the tool.

Since parsers try a vast number of rules and are inherently recursive, the logs quickly exceed $500,000$ lines for small source files. To deal with this, the logs are structured as a tree, and the user of the tool is expected to navigate this tree by selecting the precise rule attempts that the parser is supposed to succeed on, thus hiding irrelevant attempts. A dedicated lexer, in hindsight, would have reduced the exponential growth of the logs, but might not have easily parsed some of the unique language features (such as semicolons acting as both terminators and comments).

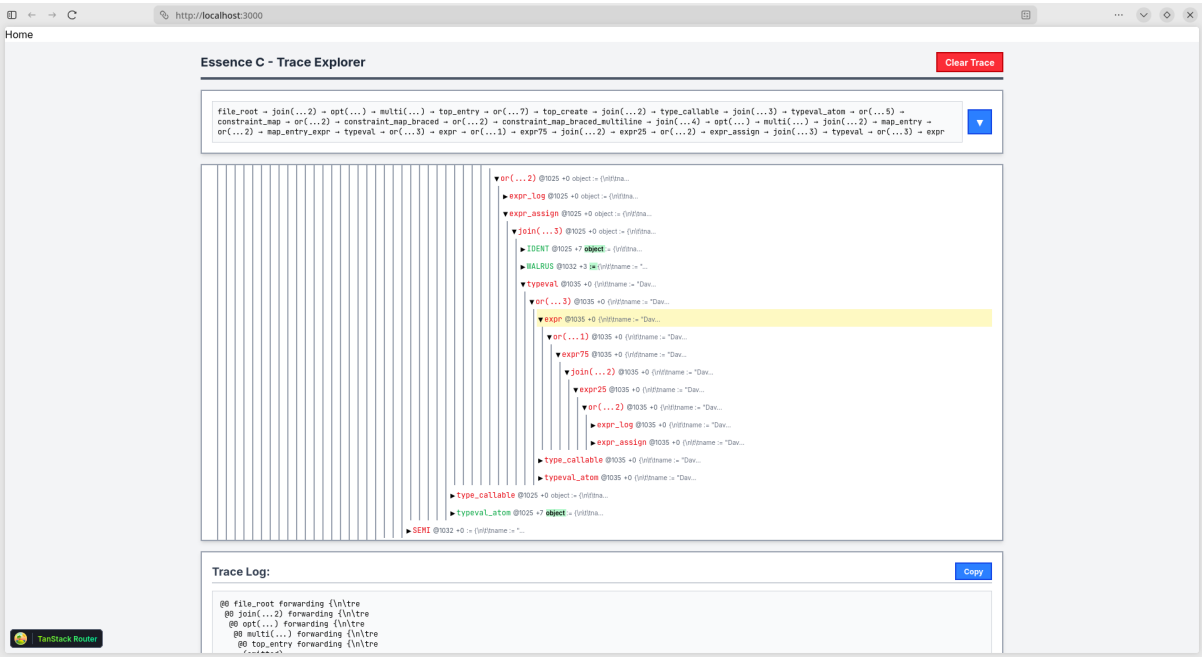

Figure 4.1: Parser Trace Explorer

4.4.3 Defensive Programming

As Essence C is designed with strict semantics, the backend has no need to recover from illegal states. As such, defensive programming is used, where the program is crashed with

a useful error message whenever an invariant is violated. Such errors could originate from semantic violations in the user's code, or due to subtle bugs in the compiler itself; either way, the issues can be traced effectively. Naturally, not all subtleties can be pre-envisioned, so this was done on a best-effort basis to avoid compromising development speed.

4.4.4 Debugging Primitives

While the primary mechanism of performing external communication is to link with external code via the C ABI (which may be LLVM IR), debugging functionality is often preferable within the language directly in case there are issues with the interoperability layer proper. Thus, two basic logging functions were implemented, "log" and "log_d". The former takes a null-terminated ascii string and prints this output in an implementation-specific manner. The latter takes an arbitrarily-sized value (like an integer or a byte array) and prints its raw binary representation (in 64-bit chunks), including both its hex representation, ascii representation, and decimal representation. The implementation of this functionality was chosen to be implementation-specific. In the prototype implementation, the external "puts" function was selected to implement this functionality, available in many environments. In case the environment does not support it, a dummy no-op implementation can be defined, effectively dropping any logs.

4.4.5 Bit-width Analysis

Determining the maximum bits needed to hold an integer is an important optimization. While LLVM may be able to detect certain constraints about numbers, much of the compiler information is lost at this point that a modern programming language might be interested in tracking. For example, software verification languages may keep track of exact min and max values for an integer type, which is tracked extensively as the type system navigates the codebase. For non-integer types, an optimal type system implementation might allow the user to use formal logic to keep track of the theoretically-smallest set of runtime possibilities (a provably accurate picture of the possible values that can be assigned at runtime, with no wasted space).

Only simple bit-tracking is performed in the prototype compiler. When a hardcoded integer constant is used, the number of bits is trivially calculated using the logarithm. Addition was more tricky to implement – one must recursively track the bits needed to hold an arbitrary sum of other integers. The first attempt of taking the maximum of the operand bits and adding one (to account for a potential carry) was insufficient for more than two operands. While an additional carry bit could be added for each additional operand, this estimate would be too conservative. The correct solution was to sum the

maximum value each operand could take on (interpreted as data to avoid to intricacies of signed integers), and then take the logarithm of this sum. This can achieved easily in languages that support arbitrarily sized integers out-of-the-box. However, it was easier to implement an approximate solution in C++ by using the double data type, with the final result rounded up. While this sounds bizarre and inaccurate at first glance, recall that floats are internally implemented using an exponent and mantissa. Thus, it is very similar to running the numbers through a logarithmic transformation, which would yield the final result directly. Furthermore, an epsilon value was included to deal with precision errors, favouring a more conservative bit estimate. It was not proven whether this overall approach is mathematically sound up to a particular bit count, but it can always be substituted if the need arises.

4.4.6 Build Environment

Early on in development, I was adamment to ensure a fully reproducible build environment. All building is done using the terminal, so that the build process is independent from the editor used. The JavaScript portion of the project uses Node.js and pnpm, which is easy to install on different operating systems, and build scripts are organized in package.json files. However, the C++ portion, consisting of a suite of tools (gcc, cmake (or is it called make), etc), is more difficult to set up. The build scripts for this portion are organized using a Makefile, and the development environment itself is setup using a Dockerfile. Some developer scripts were implemented in Bash to orchestrate the overall process between the JavaScript and C++ portions, the Docker layer, and the developer's own environment. The choice to use Bash means the developer must either use a Linux environment or a WSL environment if using Windows. A docker-in-docker strategy could theoretically have been used instead, requiring minimal platform-specific scripts (I thought I did use docker in docker, I better check), but this complexity was avoided. MacOS users may suffer as a result. Additionally, I recently switched to NixOS on my own system, and its philosophy was applicable here. NixOS allows you to write description-based setup descriptions, which can act as reproducible development enviornments. Thus I set up a repository-level nix script for those on NixOS who want to fast-track the development environment. Thus both the build layer and host interaction layer are highly reproducible.

4.4.7 Underlying Type Tracking

Chapter 5 – Evaluation

5.0.1 Manual Snapshot Testing

Throughout the development cycle, manual testing was conducted by compiling sample source files and inspecting the terminal output after running the final executable. This mirrors the snapshot testing philosophy commonly implemented in web-development projects (although accomplished in a less automated fashion here), where input/output consistency is checked before each commit. Down the road, this strategy could be extended to a full test suite that prevents all sorts of regressions. One could take this a step further by checking consistency in the compiled LLVM IR code itself, but this is typically only done when a language reaches a pivotal level of maturity; the compiler is in too much flux at this point for this technique to have any merit.

Much of the snapshot testing was performed using the language's data logging facility which prints the hexidecimal, ascii, and decimal values of arbitrarily-sized binary sequences. This allows straightforward investigation of any runtime values, whether that be integers, floats, pointers, strings, enums, and so on.

5.0.2 User Acceptance Testing

Four students from Maynooth University were tasked with implementing a program in the language. Each student was asked to provide positive and negative feedback of the solution.

The users were given the following code with no explanation, but were allowed to ask me questions.

Table 5.1: Feedback survey

Question	**Answer**
Student A	
What is the sample program doing?	[Answer]
What semantic or syntactic element of the sample program strikes you the most, and why?	[Answer]
Implement a program that does blah. Provide one point of positive feedback about the language.	[Answer]
Provide one harsh criticism about the language.	[Answer]
Student B	
What is the sample program doing?	[Answer]
What semantic or syntactic element of the sample program strikes you the most, and why?	[Answer]
Implement a program that does blah. Provide one point of positive feedback about the language.	[Answer]
Provide one harsh criticism about the language.	[Answer]
Student C	
What is the sample program doing?	[Answer]

Question	**Answer**
What semantic or syntactic element of the sample program strikes you the most, and why?	[Answer]
Implement a program that does blah. Provide one point of positive feedback about the language.	[Answer]
Provide one harsh criticism about the language.	[Answer]
Student D	
What is the sample program doing?	[Answer]
What semantic or syntactic element of the sample program strikes you the most, and why?	[Answer]
Implement a program that does blah. Provide one point of positive feedback about the language.	[Answer]
Provide one harsh criticism about the language.	[Answer]

5.0.3 Benchmarking

Chapter 6 – Conclusion

Given the time constraints for this project, the evaluation only scratched the surface. One would want to spend the following time:

- 200 more hours on this.
- 200 more hours on that.

6.0.1 Contribution to state-of-the-art

- inlining - built in language constructs to avoid likes of kernel static_call - continuum of dispatch, not just static vs dynamic, but by merging the notion of class instances (and fusing classes and modules), if we detect multiple instances we can switch to an even more dynamic dispatch mechanism. - more efficient inter-language interoperability

Chapter 7 – Parsing Library

Prior to this project I had already developed a simple parsing library in JavaScript, which allows for:

(1) Parsing terminal rules using regular expressions.

(2) Forming non-terminal rules by combining other rules in the following ways, sufficient to develop any complex grammar:

  (1) “or” making rule optional

  (2) “join” sequencing rules

  (3) “multi” repeating rules zero or more times

  (4) “opt” making rule optional

(3) Map parsed data to custom structures using the “mapData” function.

width=!,height=!, pages=1, frame, scale=0.65, pagecommand=

.1 Appendix: Interim Report

width=!,height=!, pages=2-5, frame, scale=0.75, pagecommand=